

GNet: Visual Lip Reading for German Word Classification

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Abstract

We present GNet, a visual lip-reading model for German word-level classification trained on GLips, a corpus of approximately 250,000 clips of Hessian Parliament speakers covering 500 word classes. GNet couples a 3D-CNN and ResNet-18 visual frontend with a lightweight temporal back-end consisting of a multi-scale depthwise convolutional stem followed by a two-layer Transformer encoder. It is compared against an MS-TCN back-end sharing the frontend, the data pipeline and the training recipe, so that the comparison isolates the temporal model. On the 15-word GLips benchmark GNet reaches 59.3% top-1 accuracy from scratch and 69.5% with in-domain GLips-500 to GLips-15 transfer, compared with the previously published 48.4% [1] and 50.9% from scratch or 54.8% with cross-lingual LRW pretraining [7]. We also report the first results on the full GLips vocabulary (498 classes with a complete published split), reaching 33.5% top-1 accuracy against a chance rate of 0.2%. Controlled ablations indicate that the improvement is attributable primarily to preprocessing and training data rather than to the temporal architecture: mouth-ROI preprocessing accounts for 29.2 points of top-1 accuracy, whereas the Transformer and MS-TCN back-ends are statistically indistinguishable on the 15-word task (59.3 against 59.2 from scratch). The Transformer’s advantage appears on the 498-class task, where it gains 2.4 points (5.6σ over 24,900 test clips) using $2.9\times$ fewer temporal parameters. Negative results are reported in full, including an attentive-pooling head that does not outperform mean pooling.

1. Introduction

Visual speech recognition (VSR), the classification of spoken words from silent video of a speaker’s face, has advanced rapidly on English benchmarks such as LRW, but progress in other languages remains limited by data availability. For German the only word-level benchmark is GLips [7], and published results on it cover a 15-word subset only: 50.9% top-1 accuracy training from scratch, and 54.8% when transferring from the larger English LRW corpus. No result has been reported for the full 500-word vocabulary, and no study has isolated the contribution of the temporal back-end under a fixed frontend and training recipe.

This paper addresses both gaps. We build GNet, a lip-reading model whose visual frontend follows the standard design for word-level VSR (a single spatiotemporal convolution followed by a per-frame ResNet-18) and whose temporal back-end is a multi-scale depthwise convolutional stem feeding a shallow Transformer encoder. Holding the frontend, the data pipeline, the augmentation stack and the optimiser settings fixed, we substitute only the temporal module for an MS-TCN baseline, so that reported accuracy differences are attributable to the back-end. We evaluate the same matched pair across three settings: the full vocabulary, the 15-word subset from scratch, and GLips-500 to GLips-15 in-domain transfer, with architecture and preprocessing ablations in addition.

GNet exceeds the published results on this benchmark by a substantial margin. Our ablations attribute the gain primarily to preprocessing and training data rather than to the temporal architecture. On the 15-word task the two back-ends differ by 0.13 points (0.05σ), whereas removing the mouth-ROI crop costs 29.2 points, an effect an order of magnitude larger than any back-end difference we measured. The Transformer’s advantage is confined to the full-vocabulary task, where 24,900 test clips provide sufficient statistical resolution ($+2.4$ points, 5.6σ), and it is obtained with a $2.9\times$ smaller temporal module. We therefore characterise it as a parameter-efficiency result with a large-vocabulary accuracy benefit rather than as a general accuracy advantage. Our contributions are:

- **New best published results on the GLips 15-word benchmark.** GNet reaches 59.3% top-1 accuracy from scratch, exceeding the 48.4% of Ameer et al. [1] by 10.9 points and the 50.9% from-scratch baseline of Schwiebert et al. [7] by 8.4 points, on the same corpus, the same published split and, for the former, the same word set. With in-domain transfer this rises to 69.5%, and to 70.3% for a two-back-end ensemble.
- **First full-vocabulary GLips benchmark.** We report the first results on the complete GLips vocabulary (498 of the 500 classes have a usable published split), reaching 33.5% top-1 and 53.4% top-5 accuracy against a chance rate of 0.2%, with a matched MS-TCN baseline.

- **In-domain transfer as an alternative to cross-lingual transfer.** Pretraining on the full GLips vocabulary and fine-tuning on the 15-word subset adds 10.1 points over training the same model from scratch, and exceeds the published LRW to GLips cross-lingual result by 14.7 points. The two transfer settings are not directly comparable, and we treat this as evidence that in-domain pretraining is a viable option for low-resource VSR rather than as a like-for-like comparison.
- **A controlled MS-TCN and Transformer ablation.** Both back-ends share the frontend, data, augmentation and optimiser. The Transformer gains 2.4 points on the full vocabulary (5.6σ) and ties on the 15-word task (0.05σ), using $2.9\times$ fewer temporal parameters (2.2M against 6.4M) and $1.3\times$ fewer overall (13.4M against 17.6M).
- **An accounting of the sources of the improvement, including negative results.** Ablations isolate mouth-ROI preprocessing (29.2 points), the convolutional stem (2.4 points) and the attentive-pooling head (-1.2 points, indicating no gain over mean pooling).

2. Related Work

2.1 Architectures for large-vocabulary visual speech recognition

GNet’s frontend and one of its two temporal back-ends derive from the LRW and LRW-1000 literature. Stafylakis and Tzimiropoulos [8] established the standard frontend: a single spatiotemporal convolution over short frame windows, followed by a per-frame 2D ResNet and a recurrent temporal model. Martinez et al. [5] replaced the recurrent back-end with a multi-scale temporal convolutional network (MS-TCN), simplifying training to a single stage, improving generalisation across sequence lengths, and setting state-of-the-art results on LRW and LRW-1000. This 3D-CNN, ResNet-18 and MS-TCN combination is the direct ancestor of both of GNet’s configurations. Later work extended the convolutional line: the densely connected TCN of Ma et al. [4] added skip connections between blocks to widen the temporal receptive field, and more recent multi-dilated variants match or exceed TCN accuracy at lower parameter cost [2]. Attention- and Transformer-based back-ends form a separate line, replacing the recurrent or convolutional temporal module outright [6].

GNet combines both approaches: we retain a multi-scale convolutional stem and place a shallow Transformer encoder on top of it, reporting both variants as a matched pair rather than as a single substitution. To our knowledge none of the cited work performs a controlled MS-TCN and Transformer comparison holding data, frontend and training recipe fixed; architectural changes there are evaluated against a fixed prior baseline, which confounds the back-end’s contribution with concurrent changes in the training procedure. Our results suggest this confound is material, since under a matched recipe the back-end gap is small and does not consistently favour one family.

2.2 German visual speech recognition

Schwiebert et al. [7] introduced GLips, approximately 250,000 clips of Hessian Parliament speakers over 500 word classes, formatted to match LRW so that cross-lingual transfer is possible, and showed that LRW to GLips transfer improves both convergence speed and final accuracy relative to training from scratch. This established the first German VSR baseline, but on a 15-class subset only, via cross-lingual transfer only, and without a full-vocabulary result. The reported figures are 50.9% top-1 accuracy from scratch and 54.8% with LRW pretraining.

Ameer et al. [1] benchmark a NASNetMobile-based pipeline on the same 15-class GLips subset, running a 16-configuration sweep crossing four pretrained 2D backbones with four temporal back-ends, and comparing against two further structured-dataset baselines. Their best configuration (NASNetMobile with a CNN back-end, 4.68M parameters) reaches 48.40% accuracy and 0.485 macro- F_1 without in-domain transfer.

The two pipelines differ substantially in preprocessing. Ameer et al. [1] avoid face detection, processing uncropped frames at 128×128 with image- and feature-level augmentation, on the grounds that cropping discards information relevant to unconstrained data. GNet uses MediaPipe FaceMesh [3] to extract a tracked, scale-normalised mouth crop per frame (Section 4.1). Section 5 quantifies the effect of this choice, which is the largest single factor in our results, and Section 5.5 reports the behaviour of our architecture under their input regime.

2.3 Summary of the gap addressed

No prior study on GLips, including the dataset paper, the sweep of Ameer et al. [1], or the structured-dataset baselines they cite, evaluates the full vocabulary, runs a matched MS-TCN and Transformer ablation, or compares in-domain transfer (GLips-500 to GLips-15) against the cross-lingual LRW transfer that has so far been the only transfer result reported for this corpus. GNet is narrower than Ameer et al. [1] in backbone

Table 1. Symbol table for the two temporal back-ends.

Symbol	Meaning	Value
T	clip length (frames)	25
f_t	per-frame ResNet-18 feature	$\in \mathbb{R}^{512}$
k	multi-scale kernels (both back-ends)	$\{3, 5, 7\}$
<i>GNet</i>		
d	projected / model width	256
z_t	token at time t (after MS-stem)	$\in \mathbb{R}^d$
L	Transformer layers	2
n_h	attention heads	8
d_{ff}	FFN width	1024
w_t	pooling weights, $\text{softmax}(g(z_t))$	$\sum_t w_t = 1$
<i>MS-TCN baseline</i>		
B	MS-TCN blocks	4
w	MS-TCN channel width	384
C	output classes	498 / 15

diversity and relies on landmark-tracked mouth-ROI cropping rather than their detection-free design. To our knowledge it is nonetheless the first work to vary vocabulary size, transfer setting and back-end architecture together in a controlled comparison on GLips.

3. Method

We define the visual encoder once and reuse it unchanged for both temporal back-ends, so that the comparison isolates the temporal model.

Shared visual encoder. An input clip $x \in \mathbb{R}^{3 \times T \times H \times W}$ is processed by a *Frontend3D*, a single 3D convolution with kernel $k = (5, 7, 7)$ and stride $(1, 2, 2)$ that preserves the temporal axis while halving the spatial axes, followed by batch normalisation, ReLU and a spatial max-pool. Each of the T frames is then passed independently through the ImageNet-pretrained ResNet-18 stages (**layer1** to **layer4**), yielding a per-frame feature $f_t \in \mathbb{R}^{512}$ after global spatial average pooling. GLips clips are 1.16s at 25 fps; we sample $T = 25$ frames per clip at $H = W = 88$. Both back-ends consume the same feature sequence $F \in \mathbb{R}^{T \times 512}$, and the encoder accounts for 11.2M of each model’s parameters.

Temporal back-end (GNet). F passes through (i) a linear projection $\text{Linear}(512 \rightarrow d) + \text{LayerNorm}$ with $d = 256$; (ii) a multi-scale convolutional stem of two **MSTemporalBlocks**, each averaging three parallel depthwise branches with kernels $k \in \{3, 5, 7\}$, where each branch comprises a depthwise convolution, a pointwise convolution, batch normalisation and GELU, under a residual connection and LayerNorm; (iii) a learned positional embedding; (iv) a 2-layer pre-norm Transformer encoder (8 heads, FFN width 1024, GELU, dropout 0.2); and (v) attentive temporal pooling followed by dropout and a linear classifier. Attentive pooling scores each token with a small two-layer network g , forms weights $w_t = \text{softmax}(g(z_t))$, and returns $\sum_t w_t z_t$. The temporal module totals 2.2M parameters, giving 13.4M for the whole model on the 15-class task.

Temporal back-end (MS-TCN baseline). The same F is fed to a multi-scale temporal convolutional network operating directly on the 512-d sequence. It stacks $B = 4$ MS-TCN blocks of width 384; each block runs parallel 1-D temporal convolutions with kernels $k \in \{3, 5, 7\}$ at increasing dilation, followed by batch normalisation, ReLU and a residual connection. A final temporal average pool over T feeds a linear classifier. With no attention but wider channels, the baseline’s temporal module uses 6.4M parameters against 2.2M for GNet, and 17.6M against 13.4M overall.

3.1 Design rationale

Because both back-ends share the encoder, any accuracy gap is attributable to the temporal model. The frontend is the standard design for word-level VSR, a single spatiotemporal convolution followed by a per-frame ResNet, as introduced by Stafylakis and Tzimiropoulos [8] and reused by subsequent LRW systems [5, 4], so reusing it unchanged is what makes the comparison an ablation of the back-end.

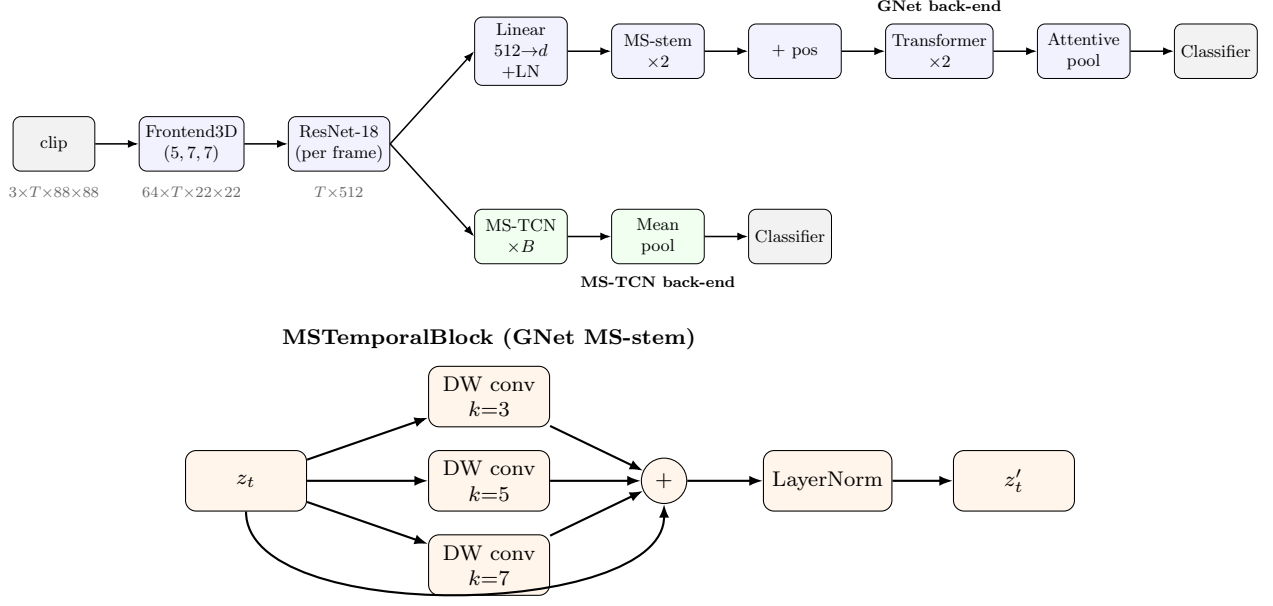


Figure 1. Shared visual encoder feeding the two temporal back-ends (top): the GNet path (projection, multi-scale stem, Transformer, attentive pooling) and the MS-TCN baseline (stacked dilated temporal-conv blocks, mean pooling). One **MSTemporalBlock** of the GNet MS-stem is detailed below; the three branch outputs are averaged, not summed, before the residual connection. Shapes assume $T=25$ and the stride-2 convolution followed by the stride-2 max-pool of the Frontend3D.

Temporal stride 1 in the Frontend3D. The frontend convolves over short windows (kernel $k = (5, 7, 7)$) at unit temporal stride, so that no frames are merged before the back-end, preserving the short-range articulatory dynamics that separate visually similar words. The same short-window, unit-stride design underlies the canonical LRW frontend [8] and the sub-clip encoders used in more recent attention-based systems [6].

Multi-scale depthwise branches ($k \in \{3, 5, 7\}$). Multi-scale temporal convolution is the basis of the TCN back-end’s competitiveness: Martinez et al. [5] improved over the recurrent baseline by mixing short and long temporal receptive fields, and dense or dilated successors [4, 2] indicate that broad receptive-field coverage, rather than depth, is the principal accuracy lever, and that it can be obtained at low cost. Our depthwise branches are an inexpensive realisation of this principle; Section 5 measures their contribution at 2.4 points of top-1 accuracy.

Shallow (2-layer) Transformer. GLiPs provides 500 clips per class [7], that is, 400 training clips per word under the published split, a regime in which self-attention overfits readily. We cap the stack at 2 layers, reduced from an initial 4 which overfit measurably, and rely on dropout.

Attentive pooling. Each clip is largely non-articulatory, since the target word occupies only a fraction of the 25 frames, so a uniform temporal average dilutes the informative frames, and query-based attentive pooling is intended to up-weight them [6]. The ablation in Section 5 does not support this rationale: a mean-pooling variant performs equivalently. We retain attentive pooling as the default configuration because the transfer runs were trained with it.

Transfer procedure. GLiPs is designed to be LRW-compatible, and Schwiebert et al. [7] show that transfer between the two corpora accelerates convergence and improves accuracy. We apply the same mechanism within GLiPs. For both back-ends, transfer warm-starts every tensor of the feature extractor whose name and shape match, comprising the 3D frontend, the ResNet, the temporal module, the positional embedding and the pooling head, and reinitialises only the classifier, so that a model with a different class count warm-starts cleanly (`load_transfer_weights`). The source is always the corresponding full-vocabulary model of the same back-end family, so the transfer comparison remains matched.

4. Experimental Setup

4.1 Dataset, splits and preprocessing

GLips [7] ships with a fixed per-class partition into *train*, *val* and *test* folders holding 400, 50 and 50 clips per word class respectively. We train on *train*, select checkpoints on *val*, and report all headline numbers on the held-out *test* folder. We use the published folder split verbatim, with no source-disjoint re-partitioning, so that our numbers are comparable to prior work adopting the same split. On the 15-class benchmark this yields 6,000 training, 750 validation and 750 test clips. For the full-vocabulary task we use the 498 classes that have a complete published split; two classes (*soll*, *hier*) are degenerate in the corpus, having no validation folder, and are dropped so that train and validation cover an identical class set. This gives approximately 199,000 training and 24,900 test clips.

For the 15-word subset we adopt the word set of Ameer et al. [1]: *aber*, *aufgaben*, *bleibt*, *darüber*, *digitalisierung*, *einmal*, *finden*, *gegen*, *geworden*, *herrn*, *investitionen*, *kommunen*, *linke*, *möglichkeit*, *passiert*. We do not use the first fifteen words alphabetically, which are dominated by short, visually similar function words and cap out near 28% accuracy in our own runs. A randomly drawn subset would make our results incomparable to prior work, since accuracy depends strongly on which words are drawn, and exhaustive evaluation is infeasible, there being $\binom{500}{15} \approx 1.88 \times 10^{28}$ possible draws. Schwiebert et al. [7] do not report their 15-word draw, so our comparison against their figures is approximate; the comparison against Ameer et al. [1] is on their published word set.

Mouth-ROI pipeline. For every frame, MediaPipe FaceMesh [3] localises the lip landmarks. An axis-aligned square box is fitted around the lip region with side length $s = \text{box_scale} \times \max(w_{\text{mouth}}, h_{\text{mouth}})$ at $\text{box_scale} = 1.6$, exponentially smoothed across frames ($\alpha = 0.4$) and held over brief detection dropouts. The box is cropped from the native 256×256 frame and resized to 96×96 , and the original audio is re-muxed so that multimodal variants retain sound. At training time the 96×96 crop is randomly cropped to 88×88 (centre crop at evaluation), 25 frames are temporally sampled per clip, and the augmentation stack of Section 4.2 is applied with ImageNet channel normalisation.

Baseline-matched pipeline. To separate preprocessing from architecture we also reproduce the input recipe of Ameer et al. [1] on the uncropped frames: no face detection or mouth-ROI crop, each frame resized directly to 128×128 with no spatial crop, 16 frames sampled per clip, per-clip min-max normalisation to $[0, 1]$, and random horizontal flipping as the only augmentation. Holding the model and optimiser fixed while switching only between these two pipelines isolates the contribution of the mouth-ROI preprocessing from that of the architecture.

4.2 Data augmentation

Lip reading depends on the temporal evolution of mouth shape, so our augmentation is built around one invariant: every stochastic transform parameter is drawn once per clip and applied identically to all T frames. A spatial jitter, colour shift or occlusion varying from frame to frame would inject apparent motion unrelated to articulation. All augmentation is active during training only; at validation and test the transform reduces to a deterministic resize, centre crop and normalisation.

Geometric. A single affine warp per clip, with rotation drawn uniformly from $[-10^\circ, 10^\circ]$, isotropic scale from $[0.9, 1.1]$, and translation up to 6% of the frame along each axis, followed by a random 88×88 crop (centre crop at evaluation) and a horizontal flip with probability 0.5. The affine is applied before the crop so that the crop discards most of the border region the warp introduces.

Photometric. Per-clip brightness and contrast jitter, each a multiplicative factor in $[0.8, 1.2]$, and random conversion of the whole clip to grayscale with probability 0.2. Grayscale removes a skin-tone and lighting shortcut: because lip reading depends on shape rather than colour, removing the colour channel on a fifth of the clips discourages memorisation of speaker- and studio-specific appearance.

Temporal. Two mechanisms perturb the time axis while holding the frame count fixed at 25. A speed perturbation resamples the 25 frames from a randomly warped and offset sub-window of the clip, corresponding to an effective speaking-rate factor in $[0.8, 1.2]$, which augments articulation speed and adds a temporal crop. Temporal masking then zeros between one and two contiguous spans of up to 4 frames each, encouraging the temporal back-end to integrate over the whole utterance rather than rely on a few decisive frames.

Occlusion. With probability 0.25 we apply random erasing: a single rectangle covering 2 to 20% of the frame area at aspect ratio in $[0.3, 3.3]$ is filled with a flat value. The same rectangle is erased in every frame, consistent with the clip-invariance principle above, simulating partial static occlusion of the mouth.

Table 2. Test-split results on the published GLips split. Parameter counts correspond to the class count of the row. Chance is 0.2% on GLips-498 and 6.7% on GLips-15.

Task	Back-end	Params	Top-1	Top-5	Macro- F_1
GLips-498	MS-TCN	17.8M	31.18	50.51	29.28
	GNet	13.6M	33.53	53.37	32.02
GLips-15 (scratch)	MS-TCN	17.6M	59.20	86.67	58.09
	GNet	13.4M	59.33	85.73	58.23
GLips-15 (transfer)	MS-TCN	17.6M	68.53	89.47	67.67
	GNet	13.4M	69.47	88.93	69.21
GLips-15 (transfer)	Ensemble	31.0M	70.27	90.00	69.80

4.3 Training setup

All models are optimised with AdamW (weight decay 0.05) under a differential learning rate: the ImageNet-pretrained ResNet-18 backbone is fine-tuned at $\eta_{bb} = 10^{-4}$, while all freshly initialised parameters (the 3D stem, the temporal module, the positional embedding, the pooling head and the classifier) are trained at $\eta_{head} = 10^{-3}$. The schedule is a 3-epoch linear warm-up from 0.1η to η followed by cosine annealing. We use batch size 32, gradient-norm clipping at 1.0, and bfloat16 automatic mixed precision, falling back to fp16 with loss scaling where bf16 is unavailable. Every configuration in a given comparison uses identical settings, with the component under test as the only variable.

The 15-class runs use the full regularisation recipe: label smoothing ($\varepsilon = 0.1$), batch-level Mixup and CutMix, the augmentation stack of Section 4.2, and an exponential moving average (EMA) of the weights with decay 0.999. All validation and checkpoint selection use the EMA weights, and the reported checkpoint is the EMA snapshot at peak validation top-1 accuracy. These runs are budgeted at 80 epochs with early stopping on validation top-1 accuracy. The full-vocabulary runs, which serve as transfer feature sources (Section 3.1) and whose classifiers are subsequently discarded, use a plain recipe without EMA or Mixup and are trained for 30 epochs, which is sufficient for the frontend and temporal back-end to converge as a feature extractor and keeps full-vocabulary training tractable on our hardware. All experiments run on a single NVIDIA RTX 3060 (12 GB).

4.4 Evaluation metrics

We report top-1 and top-5 classification accuracy and macro-averaged F_1 . For N clips with predicted logits $\hat{y}_i$ and ground-truth label y_i , top- k accuracy is

$$\text{Top-}k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i \in \text{top-}k(\hat{y}_i)],$$

where $\text{top-}k(\cdot)$ returns the indices of the k largest logits, with $k = \min(5, C)$ for a C -class task. Top-1 is the primary metric for model selection and all head-to-head comparisons; top-5 characterises near-miss behaviour on a visually confusable vocabulary; macro- F_1 is reported because it is the secondary metric used by Ameer et al. [1] and because it is insensitive to the mild class-count imbalance introduced by dropping degenerate classes. Every row of every results table is scored by a single shared evaluation script under one protocol, with the same split, class list and batching, applying each row’s own preprocessing.

5. Results

All numbers in this section are top-1 and top-5 accuracy and macro- F_1 on the held-out GLips *test* folder. Where we compare two of our own configurations we also quote the significance of the gap, computed as $|\Delta|/\sqrt{\text{SE}_a^2 + \text{SE}_b^2}$ with $\text{SE} = \sqrt{p(1-p)/N}$. This is relevant here because the 15-word test split contains only $N = 750$ clips, so differences below approximately 5 points are not resolvable on it. The full-vocabulary split, at $N = 24,900$, resolves differences approximately six times finer.

5.1 Main comparison

Table 2 gives the main comparison, from which three observations follow.

On the full vocabulary the Transformer back-end is ahead. GNet reaches 33.53% top-1 accuracy against MS-TCN’s 31.18%, a gap of 2.35 points at 5.6σ over 24,900 test clips, with a corresponding 2.7-point advantage

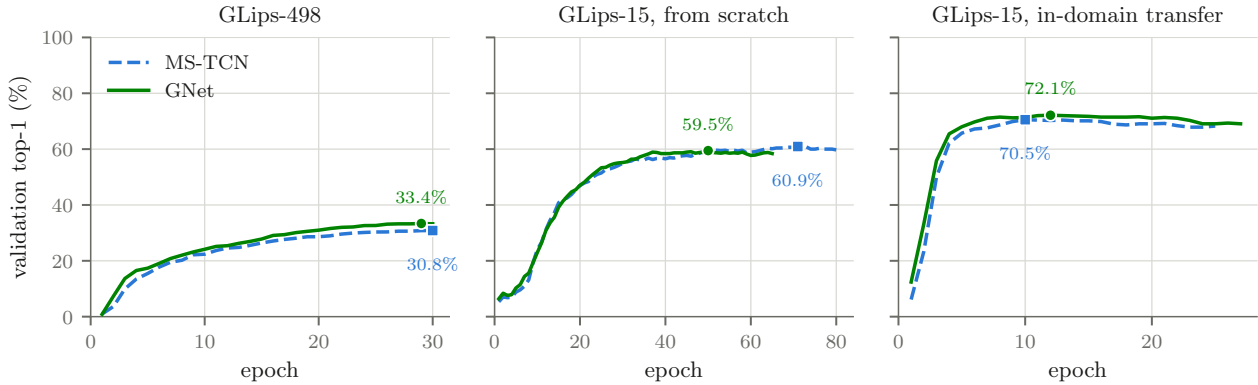


Figure 2. Validation top-1 accuracy against epoch for the two back-ends in each setting. The two curves are close throughout on both 15-word panels, consistent with the test-split result that the back-ends are statistically indistinguishable at this vocabulary size, and separate steadily on GLips-498. The transfer runs (right) start above 12% at epoch 1 and peak within 12 epochs, against roughly 50 for the same architecture trained from scratch, reproducing within GLips the convergence-speed benefit Schwiebert et al. [7] report for LRW to GLips transfer. Curves are validation accuracy under EMA weights on the 15-word panels; the reported headline numbers are test-split and appear in Table 2.

Table 3. GLips 15-word benchmark against published results. All rows use the corpus’s published split. Ameer et al. [1] use the same word set; Schwiebert et al. [7] do not publish their draw, so that comparison is approximate.

Method	Pretraining	Top-1	Macro- F_1
Schwiebert et al. [7]	none (from scratch)	50.9	–
Ameer et al. [1] (NASNetMobile + CNN)	ImageNet	48.40	48.50
Schwiebert et al. [7]	LRW (cross-lingual)	54.8	–
GNet (ours)	ImageNet	59.33	58.23
GNet (ours)	ImageNet + GLips-498	69.47	69.21
GNet + MS-TCN ensemble (ours)	ImageNet + GLips-498	70.27	69.80

in macro- F_1 , and obtains it with 4.2M fewer parameters. This is the only architectural difference in our study that lies clearly outside the noise floor. Both figures are, to our knowledge, the first reported for the full GLips vocabulary; against a chance rate of 0.2%, 33.5% top-1 and 53.4% top-5 accuracy indicate that the task is learnable but far from saturated.

On the 15-word task the two back-ends are indistinguishable. From scratch they differ by 0.13 points (0.05σ), and under transfer by 0.94 points (0.39σ). Neither gap is resolvable at $N = 750$, and the top-5 column favours MS-TCN in both rows. We therefore do not claim a Transformer accuracy advantage at this vocabulary size. GNet does, however, reach the same accuracy with a $2.9\times$ smaller temporal module.

In-domain transfer contributes more than the back-end choice. Warm-starting from the full-vocabulary model adds 10.1 points for GNet (59.33 to 69.47) and 9.3 points for MS-TCN (59.20 to 68.53), an effect an order of magnitude larger than any architectural difference at this scale. A softmax-average ensemble of the two transfer models adds a further 0.8 points, reaching 70.27%; we report it for completeness, noting that it doubles inference cost for a sub- σ gain.

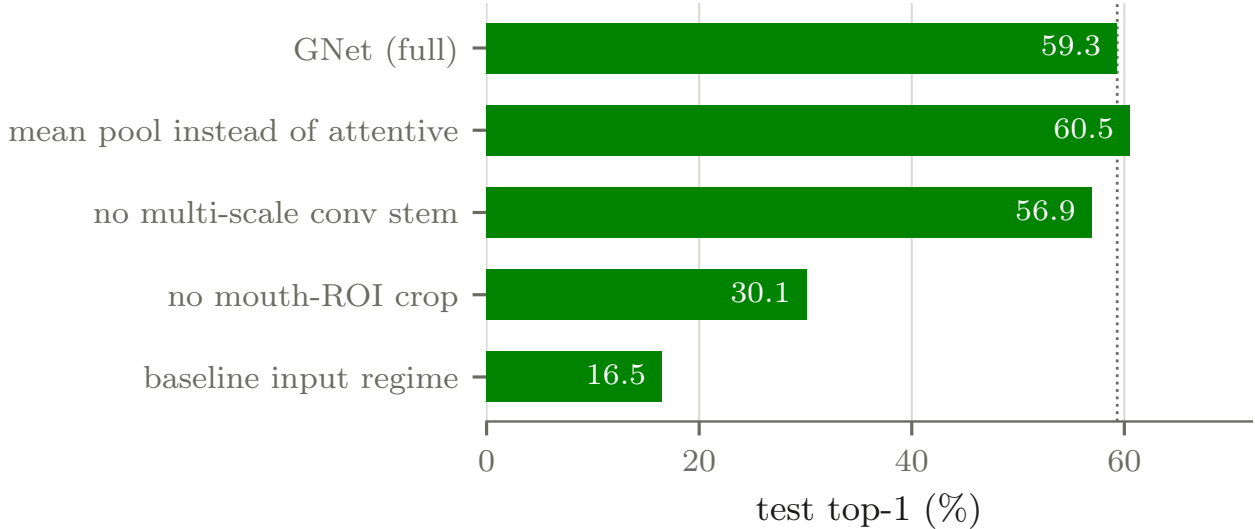
Figure 2 shows the validation curves underlying these results. Two features are visible beyond the final accuracies. The back-end curves overlap for most of training in both 15-word panels and separate only on GLips-498, which matches the significance analysis above. The transfer runs also converge considerably faster than the from-scratch runs, peaking within 12 epochs rather than roughly 50, so in-domain pretraining reduces training cost on the target task in addition to improving final accuracy.

5.2 Comparison with published results

Table 3 places our results against the published literature. The closest comparison is the from-scratch row: with only ImageNet initialisation, as in Ameer et al. [1], GNet reaches 59.33% against their 48.40%, an improvement of 10.9 points on the same corpus, the same 15 word classes and the same published split, with a 9.7-point macro- F_1 margin. Adding in-domain GLips-498 pretraining, which uses only GLips training data but more of it than the 15-class training set alone, raises this to 69.47%, exceeding the 54.8% that Schwiebert et al. [7] obtain by transferring from the larger English LRW corpus.

Table 4. GNet ablations on GLips-15, test split. Each row is the same architecture and training recipe with a single component changed. Δ is top-1 accuracy relative to the full model, with its significance at $N = 750$.

Configuration	Top-1	Top-5	Macro- F_1	Δ	Sig.
GNet (full)	59.33	85.73	58.23	–	–
– attentive pool (mean pool)	60.53	85.33	60.06	+1.20	0.5σ
– multi-scale conv stem	56.93	83.73	55.87	–2.40	0.9σ
– mouth-ROI crop	30.13	68.13	29.38	–29.20	11.9σ

**Figure 3.** Test top-1 accuracy for each ablated configuration on GLips-15. The dotted line marks the full model. The two input-pipeline rows, removal of the mouth-ROI crop and the baseline input regime of Section 5.5, account for far larger changes than either architectural component.

Two qualifications apply. First, Schwiebert et al. [7] do not report which fifteen words they used, and word-subset difficulty varies considerably, since the first fifteen words alphabetically cap out near 28% in our own runs, so the comparison against their figures is approximate in a way that the comparison against Ameer et al. [1] is not. Second, in-domain and cross-lingual transfer are not equivalent settings: our source and target share recording conditions, speakers and language, whereas LRW to GLips transfer must additionally bridge a cross-lingual and cross-corpus gap. The 14.7-point margin is therefore evidence that in-domain pretraining is a viable alternative for low-resource VSR rather than a like-for-like comparison against cross-lingual transfer.

5.3 Ablations

Mouth-ROI preprocessing is the dominant factor. Training the identical model on uncropped full-face frames reduces top-1 accuracy from 59.33% to 30.13%, a loss of 29.2 points at 11.9σ , and the only effect in our study larger than a few points. It is approximately $12\times$ the size of the largest back-end difference we measured. This indicates that landmark-tracked mouth cropping contributes more on GLips than any temporal-architecture decision evaluated here, and that a corpus of 500 clips per class does not provide enough data for the model to learn to attend to the mouth region without an explicit cropping stage.

The convolutional stem contributes modestly. Removing it and feeding per-frame features directly into the Transformer costs 2.4 points. The direction is consistent with the multi-scale receptive-field argument of Martinez et al. [5] and we retain the stem, but at 0.9σ on this split the effect is not established and would require additional seeds or the full-vocabulary split to confirm.

Attentive pooling does not outperform mean pooling. A plain mean-pooling variant scores 1.2 points higher in top-1 accuracy and 1.8 points higher in macro- F_1 . The difference is not significant (0.5σ), so the two are best regarded as equivalent, but the design rationale in Section 3, that attentive pooling should up-weight the articulated frames in an otherwise silent clip, is not supported. One possible explanation is that the augmentation stack already enforces temporal integration, since random temporal masking zeroes short spans specifically to prevent the model from relying on a few decisive frames.

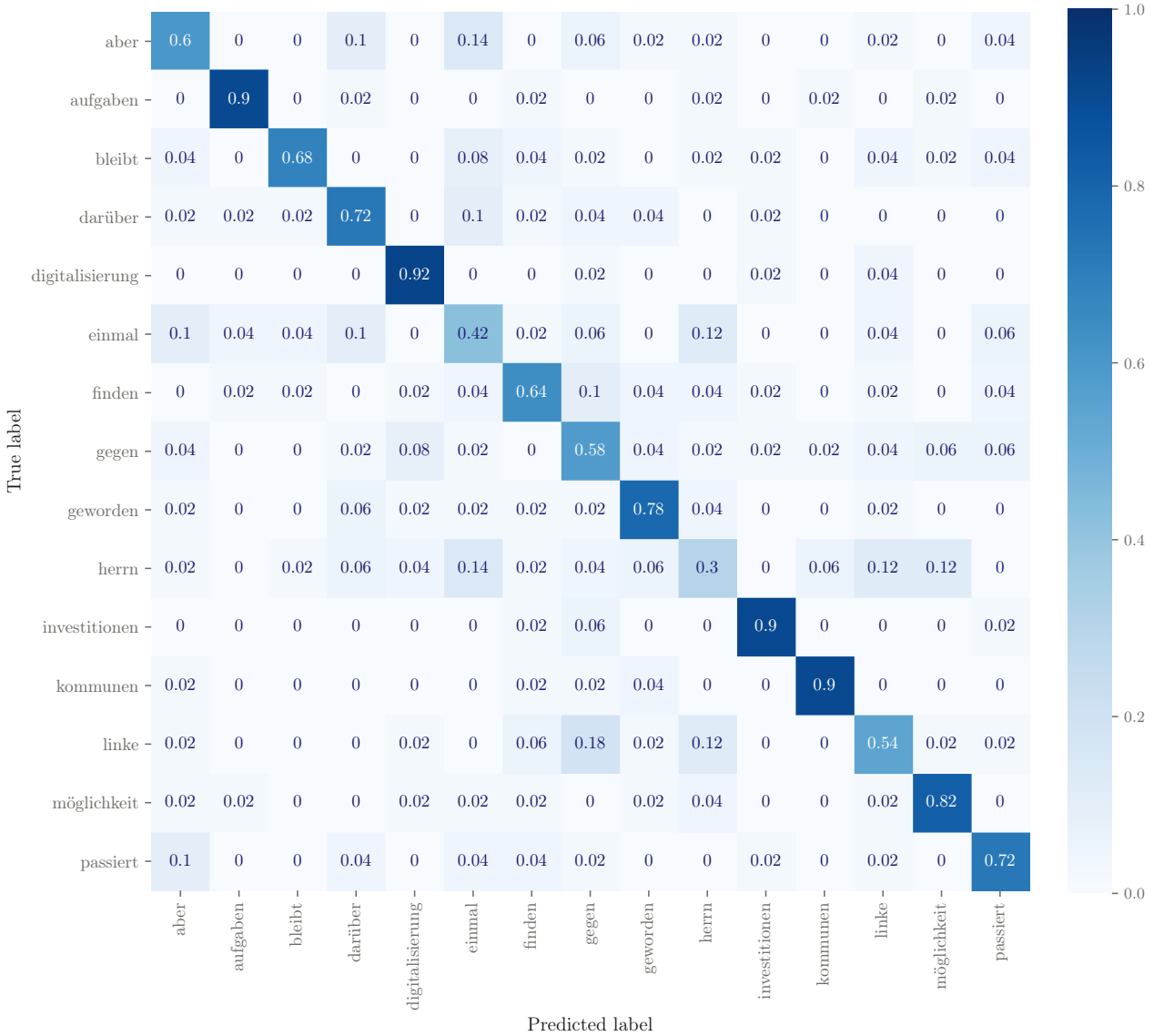


Figure 4. Row-normalised confusion matrix for GNet with in-domain transfer on the GLips-15 test split. Cells give the rate per true label, so the diagonal is per-class recall. Recall is highest for long, morphologically distinctive words and lowest for short function words, which also account for the largest off-diagonal masses.

5.4 Per-class behaviour

Figure 4 gives the row-normalised confusion matrix for the best single model. Per-class recall varies widely, from 92% for *digitalisierung* and 90% for *aufgaben*, *investitionen* and *kommunen*, down to 30% for *herrn* and 42% for *einmal*. Recall correlates strongly with orthographic word length (Pearson $r = 0.74$): classes of at least nine characters average 88.0% recall, while those of six characters or fewer average 53.7%. The largest individual confusions are *linke* predicted as *gegen* (18%), and *herrn* and *aber* both predicted as *einmal* (14% each). This is consistent with the visual-phonetic structure of the task rather than with a defect of the model: short words span fewer frames and offer fewer distinguishing visemes, and several of the low-recall classes are near-homophenes that are not separable from the lip channel alone. It also indicates that aggregate accuracy on a 15-word subset depends substantially on the length distribution of the words drawn, which is a further reason that comparisons across different word draws, such as the one against Schwiebert et al. [7], should be treated as approximate.

5.5 Behaviour under the baseline input regime

To bound how much of the improvement is architectural, we also trained both back-ends under the input regime of Ameer et al. [1]: uncropped 128×128 frames, 16 frames per clip, min-max normalisation and horizontal flipping as the only augmentation. GNet reaches 16.53% top-1 accuracy and MS-TCN 20.93%, both well below their published 48.40%.

This experiment characterises our architecture under their input regime and is not a reproduction of their method, so it should not be read as a head-to-head comparison. The two designs differ in how they use the frontend. Theirs is a 4.68M-parameter ImageNet-pretrained 2D CNN used as a feature extractor, with augmentation applied at the feature level and a light temporal head trained on top. Ours is a 13.4M-parameter network with a 3D convolutional stem, trained end-to-end with the ResNet backbone fine-tuned. Without the mouth-ROI crop and without clip-level augmentation beyond flipping, our model overfits severely, reaching 75% training accuracy against 17% on validation, which is the expected failure mode for an end-to-end model of this capacity on 6,000 clips of uncropped faces. The result is consistent with the ablation above: GNet’s accuracy is contingent on the ROI crop and the augmentation stack.

6. Discussion and limitations

The improvement is data-driven rather than architectural. GNet’s 10.9-point margin over Ameer et al. [1] is a benchmark improvement on a shared split and word set, but our ablations attribute it primarily to preprocessing and augmentation rather than to the temporal back-end. A 29-point preprocessing effect that exceeds every architectural difference measured here suggests that reported architectural gains on small VSR corpora are difficult to interpret unless preprocessing is held fixed across the compared systems.

The Transformer’s benefit is limited in scope. We observe it only at 498 classes. This is consistent with the expectation that self-attention and long-range temporal context become useful as the label space grows and finer temporal discrimination is required, whereas at 15 classes a multi-scale convolutional back-end already captures what the frontend provides. On accuracy grounds we do not find a basis for preferring the Transformer back-end over MS-TCN for small-vocabulary tasks; on parameter-efficiency grounds it is preferable at any size, and on both grounds for large vocabularies.

Statistical power is the principal methodological limitation. The published 15-word test split contains 750 clips, which resolves differences of approximately 5 points. Several ablations, and both 15-word back-end comparisons, fall below that threshold and are marked as such. All results are single-seed; a multi-seed protocol with confidence intervals would be required to settle the sub- σ comparisons, and was outside the compute budget available on a single RTX 3060.

Further limitations. We use the published split, which is not speaker-disjoint, so our numbers, like those of Ameer et al. [1] and Schwiebert et al. [7] on the same split, include speaker overlap between train and test and should be read as within-corpus rather than speaker-independent performance. GLips consists of parliamentary footage: front-facing, well-lit, adult and broadcast-quality. Our reliance on mouth-ROI cropping suits that setting and may transfer poorly to unconstrained video where face detection fails, which is the concern Ameer et al. [1] raise for their detection-free design, and Section 5.5 provides no evidence against it. Finally, the full-vocabulary models were trained for 30 epochs under a plain recipe, since they serve as transfer sources, so the 33.5% figure should be read as an initial baseline rather than a ceiling.

7. Conclusion

GNet is a German visual lip-reading system that improves on the published results for the GLips 15-word benchmark, reaching 59.3% top-1 accuracy from scratch against a previous best of 50.9% from scratch and 48.4% for the most recent published pipeline, rising to 69.5% with in-domain GLips-498 transfer and 70.3% with a two-back-end ensemble, where the previous best transfer result was 54.8% using cross-lingual LRW pretraining. We additionally report the first results on the full GLips vocabulary, at 33.5% top-1 and 53.4% top-5 accuracy across 498 classes against a chance rate of 0.2%.

Controlled ablations attribute the improvement primarily to data and preprocessing rather than to the temporal architecture, with 29.2 points from mouth-ROI preprocessing and 10.1 points from in-domain transfer, while the Transformer and MS-TCN back-ends are statistically indistinguishable at this vocabulary size. For the temporal back-end, the Transformer is the preferable default: it is ahead on the full 498-class vocabulary by 2.4 points (5.6σ) while using $2.9\times$ fewer temporal parameters, and ties elsewhere. This constitutes a parameter-efficiency result with a large-vocabulary accuracy benefit rather than a general accuracy advantage. We also report that the attentive-pooling head, adopted on the reasoning that most frames in a clip are non-articulatory, does not outperform mean pooling.

The direction these results suggest is not a different temporal module but more data and better-targeted input: a speaker-disjoint German benchmark large enough to resolve the differences this one cannot, and self-supervised audio-visual pretraining on German at a scale where the frontend can learn to localise the mouth without an explicit cropping stage.

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